Objects, Inheritance, and Linked Lists Meta

Computer Science Mentors 61A

October 21 – October 24, 2025

Example Timeline

- Ice Breaker [3 min]

  If you want, start off with an icebreaker of your choice!

  Also, there is a link to a feedback form at the bottom of this worksheet! It would be amazing if you could highlight this/fill it out yourself so we can improve the worksheets in the future!
- Mini-lecture: Inheritance and Repr/Str [7 min]
- Q1: Birds [7 min]
- Q2: Star Wars [7 min]
- Mini-lecture: Linked Lists [7 min]
- Q3: WWPD Linked Lists [5 min]
- Q4: Reverse [7 min]
- Q5: Insert At [7 min]

1 Inheritance and Representation

Created by Anaisha Das and Tavisha Patel. Worksheet feedback: forms.gle/q67BZrHHfoKbF6V28

1. **Flying the cOOP** What would Python display? Write the result of executing the code and the prompts below. If a function is returned, write "Function". If nothing is returned, write "Nothing". If an error occurs, write "Error".
Hint: You may find it helpful to make an environment diagram tracking your objects and *each* new instance of an object (whether it's assigned to a variable or not!).

```
class Bird:
    def __init__(self, call):
        self.call = call
        self.can_fly = True
    def fly(self):
        if self.can_fly:
            return "Don't stop me
                now!"
        else:
            return "Ground control to
                Major Tom..."
    def speak(self):
        print(self.call)

class Chicken(Bird):
    def speak(self, other):
        Bird.speak(self)
        other.speak()

class Penguin(Bird):
    can_fly = False
    def speak(self):
        call = "Ice to see you"
        print(call)

andre = Chicken("cluck")
gunter = Penguin("noot")
```

```
>>> andre.speak(Bird("coo"))
```

cluck
coo

```
>>> andre.speak()
```

Error

```
>>> gunter.fly()
```

"Don't stop me now!"

```
>>> andre.speak(gunter)
```

cluck
Ice to see you

```
>>> Bird.speak(gunter)
```

noot

Explanations

- `andre.speak(Bird("coo"))`

The `Bird` object is created and immediately passed in as the parameter for `Bird`. Even though we don't assign it to a variable, the object still exists and has all the features of a `Bird` object.

This might be difficult to conceptualize to students, so stop to make sure students understand how this works, since this same create-and-use method will be used often in midterm / exam level environment diagram questions.

- `andre.speak()`

Python expects two parameters but in this case we are only assigning `self`.

It might be good to emphasize to students that this is how regular functions work as well. We have to pass in the correct amount of values.

- `gunter.fly()`

Note that the `Penguin` class will use the constructor for the `Bird` class, which sets `gunter.can_fly` for the particular instance.

Step slowly through this part to emphasize how the `Penguin` class variable differs from the `gunter` instance variable for `can_fly`.

- `andre.speak(gunter)`

This question is really similar to the first one, but instead of `Bird("coo")` we use the `gunter` object instead.

- `Bird.speak(gunter)`

`Bird.speak` looks within the `Bird` class to find the speak method.

Some students may think that `Bird` is passed in for `self`. This is not the case because `Bird` is not an instance; it's a class.

Suggested Time: 7 min Feel free to skip this question if you had enough time to do the inheritance question on the last worksheet.

2. What would Python display? Write the result of executing the following code and prompts. If nothing would happen, write "Nothing". If an error occurs, write "Error".

```
class ForceWielder():
    force = 25

    def __init__(self, name):
        self.name = name

    def train(self, other):
        other.force += self.force / 5

    def __str__(self):
        return self.name

class Jedi(ForceWielder):
    lightsaber = "blue"

    def __str__(self):
        return "Jedi " + self.name

    def __repr__(self):
        return f"Jedi({repr(self.name)})"

class Sith(ForceWielder):
    lightsaber = "red"
    num_sith = 0

    def __init__(self, name):
        super().__init__(name)
        Sith.num_sith += 1
        if self.num_sith != 2:
            print("Two there should be. No more, no less.")

    def __str__(self):
        return "Darth " + self.name

    def __repr__(self):
        return f"Sith({repr(self.name)})"
```

```
>>> anakin = Jedi("Anakin")
>>> anakin.lightsaber, anakin.force
```

```
("blue", 25)
```

```
>>> obiwan = Jedi("Obi-wan")
>>> anakin.master = obiwan
>>> anakin.master
```

```
Jedi("Obi-wan")
```

```
>>> Jedi.master
```

```
AttributeError
```

```
>>> obiwan.force += anakin.force
>>> obiwan.force, anakin.force
```

```
(50, 25)
```

```
>>> obiwan.train(anakin)
>>> obiwan.force, anakin.force
```

```
(50, 35.0)
```

```
>>> Jedi.train(obiwan, anakin)
>>> obiwan.force, anakin.force
```

```
(50, 45.0)
```

```
>>> sidious = Sith("Sidious")
```

```
Two there should be. No more, no less.
```

```
>>> ForceWielder.train(sidious, anakin)
>>> anakin.lightsaber = "red"
>>> anakin.lightsaber, anakin.force
```

```
("red", 50.0)
```

```
>>> Jedi.lightsaber
```

```
"blue"

>>> print(Sith("Vader"), Sith("Maul").num_sith)

Two there should be. No more, no less.
Darth Vader 3

>>> rey = ForceWielder("Rey")
>>> rey

<__main__.ForceWielder object>

>>> rey.lightsaber

AttributeError
```

To address a point brought up in previous semesters: You may realize that as both a ForceWielder and Jedi, `anakin` has no attribute named `master` at first. Note to students that it is possible in our OOP objects for us to declare object attributes on the fly. Also communicate to students that while you can declare object attributes on the fly, that does not mean that such attributes persist to the class the object exists in as a whole.
In my opinion, going through an example like this is far more helpful for students than a mini-lecture. Try to foresee some questions and confusions might have and how you might address them, for example:

- Why, in the `__init__` method of `Sith` can we use `self.num_sith` instead of `Sith.num_sith`? And why can't we write `self.num_sith += 1`?
- Why does evaluating `rey` give us `<__main__.ForceWielder object>`, but this is not the case when we evaluated `anakin.master`?
- What's going on with `ForceWielder.train(sidious, anakin)`?
- Can we write `Jedi.train(sidious, rey)`, even though neither `rey` nor `sidious` are Jedi?

These are also questions you could bring up if students don't ask them.

Suggested Time: 7 min

2 Linked Lists

1. What will Python output? Draw box-and-pointer diagrams along the way.

```
>>> a = Link(1, Link(2, Link(3)))
```

```
+---+---+  +---+---+  +---+---+
| 1 | --|->| 2 | --|->| 3 | / |
+---+---+  +---+---+  +---+---+
```

```
>>> a.first
```

```
1
```

```
>>> a.first = 5
```

```
+---+---+  +---+---+  +---+---+
| 5 | --|->| 2 | --|->| 3 | / |
+---+---+  +---+---+  +---+---+
```

```
>>> a.first
```

```
5
```

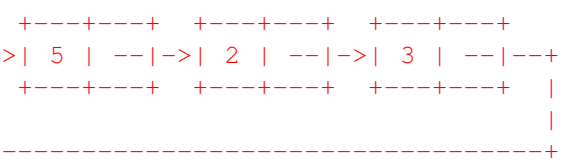
```
>>> a.rest.first
```

```
2
```

```
>>> a.rest.rest.rest.rest.first
```

Error: tuple object has no attribute rest (Link.empty has no rest)

```
>>> a.rest.rest.rest = a
```



```
>>> a.rest.rest.rest.rest.first

2

>>> repr(Link(1, Link(2, Link(3, Link.empty))))

"Link(1, Link(2, Link(3)))"

>>> Link(1, Link(2, Link(3, Link.empty)))

Link(1, Link(2, Link(3)))

>>> str(Link(1, Link(2, Link(3))))

'<1 2 3>'

>>> print(Link(Link(1), Link(2, Link(3))))

<<1> 2 3>
```

Teaching Tips

- For assignment statements, Python will not print anything but still have them draw out what the linked list will look like
- Note that we are doing mutation here, so we are actually altering the object that was created in the first assignment.
  - Some students may have minimal exposure to mutating objects so try to emphasize this and make it obvious through diagrams.
- For the error, walk-through how to keep track of which rest corresponds to which object in the box and pointer diagram. **Make sure they understand why calling rest a fourth time will give us an error (look back at the class definition)**
  - Abstraction:
    - our last .rest is set to Link.empty
    - Link.empty is not a Link objects — they do not have a .rest attribute

  - Actual implementation:
    - our last .rest is set to Link.empty
    - Link.empty is not a Link objects — they do not have a .rest attribute
- Reassigning the last .rest to point back at the front always trips students up.
  - Make it clear that a is a pointer that points to the linked list. So we are trying to assign the last rest of a to point at what a points to, which is the beginning of the list. **To test their understanding ask what would be different if we instead had**:
    - a.rest.rest.rest = a.rest
  - a way to explain the assignment for this problem is to emphasize the "evaluation" of the RHS and the LHS
  - what is the value of a (a pointer). Really emphasize the implications of pointers here.
  - where are we putting a into? (the box that represents a.rest.rest.rest)
  - same for a.rest.rest.rest = a.rest. what is the value of a.rest? (still a pointer!)
  - Mention that this creates a cycle in the list

Suggested Time: 5 min

2. Write a function `reverse`, which takes in a `Link` and returns a new `Link` that has the order of the contents reversed.
   Hint: You may want to use a helper function if you're solving this recursively.

```
def reverse(lst):
    """
    >>> a = Link(1, Link(2, Link(3)))
    >>> b = reverse(a)
    >>> b
    Link(3, Link(2, Link(1)))
    >>> a
    Link(1, Link(2, Link(3)))
    """
```

There are quite a few different methods. We have listed some here – can you think of any others?

```
# Recursive w/ Helper
def reverse(lst):
    def helper(so_far, rest):
        if rest is Link.empty:
            return so_far
        else:
            return helper(Link(rest.first, so_far), rest.rest)
    return helper(Link.empty, lst)

# Iterative
def reverse(lst):
    rev = Link.empty
    while lst is not Link.empty:
        rev = Link(lst.first, rev)
        lst = lst.rest
    return rev
```

Suggested Time: 7 min

3. Write a recursive function `insert_all` that takes as input two linked lists, `s` and `x`, and an index `index`. `insert_all` should return a new linked list with the contents of `x` inserted at index `index` of `s`.

```
def insert_all(s, x, index):
    """
    >>> insert = Link(3, Link(4))
    >>> original = Link(1, Link(2, Link(5)))
    >>> insert_all(original, insert, 2)
    Link(1, Link(2, Link(3, Link(4, Link(5)))))
    >>> start = Link(1)
    >>> insert_all(original, start, 0)
    Link(1, Link(1, Link(2, Link(5))))
    """
    if _____ and _____:

        _____

    if _____ and _____:

        _____

    _____
```

```
def insert_all(s, x, index):
    """
    >>> insert = Link(3, Link(4))
    >>> original = Link(1, Link(2, Link(5)))
    >>> insert_all(original, insert, 2)
    Link(1, Link(2, Link(3, Link(4, Link(5)))))
    >>> start = Link(1)
    >>> insert_all(original, start, 0)
    Link(1, Link(1, Link(2, Link(5))))
    >>> insert_all(original, insert, 3)
    Link(1, Link(2, Link(5, Link(3, Link(4)))))
    """
    if s is Link.empty and x is Link.empty:
        return Link.empty
    if x is not Link.empty and index == 0:
        return Link(x.first, insert_all(s, x.rest, 0))
    return Link(s.first, insert_all(s.rest, x, index - 1))
```

All of our return statements should return a new linked list.

Our base case should be the simplest possible version of the problem: when both `x` and `s` are empty, clearly the result is just the empty list.

We can now think of ways to break down this problem even further. Note that when the index to be inserted at is `0`, the problem is relatively easy: we just have to put all of the elements of `x` followed by all the elements of `s`. So the first element of the new list should `x.first`, and the rest of the new list should be `x.rest` concatenated with `s`, or `insert_all(s, x.rest, 0)`. Since we are using `x.first` and `x.rest`, we must check that `x` is nonempty to ensure that we do not error.

Finally, when the index to be inserted at is nonzero, we know that we're going to have some elements of `s`, then the elements of `x`, and then the rest of the elements from `s`. So the first element of the new list should be `s.first`. Then we can get the rest of the new list by inserting the contents of `x` at index `index - 1` of `s.rest`, reducing the index by 1 to account for the fact that we have removed the first element of `s`.

There's one issue we glossed over here: what if `x` is empty but `s` is not? Then we want to return the contents of `s`. But because the problem requires that we return a new linked list, we must recursively reconstruct `s` instead of simply returning it. You could add another base case to handle this, but as it turns out the second recursive case will handle this just fine since `Link(s.first, insert_all(s.rest, x, index - 1))` is just equivalent to `Link(s.first, s.rest)` when `x` is empty. Since the `x is not Link.empty` condition for the first recursive case will direct all situations where `x` is empty but `s` is not to the second recursive case, it turns out that we do not need to add anything else to this solution.

Convincing yourself that this problem works requires that you eventually reach a base case. Note that in either recursive call, we either reduce `s` or `x` by one element. So the base case will always eventually be reached, and the solution is valid.

Despite being just a few lines and exercising a familiar concepts with lists, I've found that this problem is quite difficult, so one thing I would emphasize is to draw out this problem with a box-and-pointer diagram and illustrate the different steps of our function. Illustrate how the function works for the doctests, which should cover all possible cases of inserting a new linked list into the beginning, middle, and end of the original linked list

If students are lost, which they most likely will be, here are some leading questions you could ask:

- When do we know that we are done inserting items into the list?
- What should the parameters be equal to if we are going to start inserting `x`, what if we are not currently inserting `x`?
- How do ensure to add all elements of `x` into `x`?

You should tell your students that they should feel free to disregard the provided skeleton, because it is quite difficult to think of the solution to this problem when you are trying to fit everything into the skeleton.

Suggested Time: 7 min This question is harder, so feel free to spend more time on it or use it as an example.